

A Survey of High-Level Synthesis-Based Hardware (IP) Watermarking Approaches

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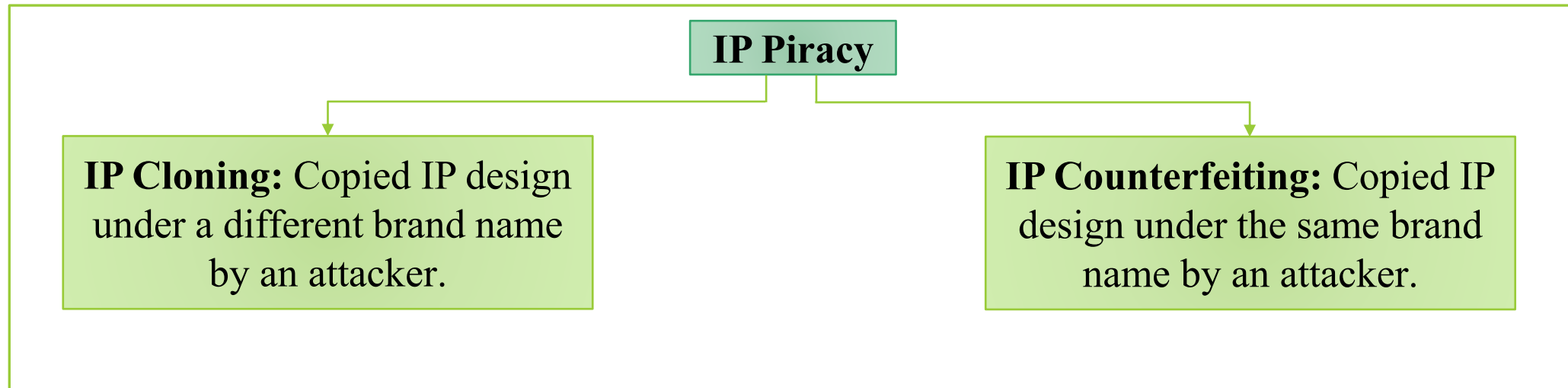
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• Introduction

- Hardware watermarking is a technique used for embedding hidden information within a hardware design.
- The demand for application-specific hardware systems is rising due to the increasing need for optimized performance, energy efficiency, and tailored solutions in various sectors.

• Threat Model

- Securing hardware designs from security threats (such as IP piracy and false IP ownership claims) is crucial in the global supply chain, necessitating robust measures such as hardware watermarking and/or hardware steganography.



• Importance of hardware watermarking

- The importance of hardware watermarking in the field of hardware IP core protection includes the following.
 - *Protection from IP piracy* – serves as a detective countermeasure.
 - *Enhancing design integrity* – making it difficult for malicious actors.
 - *Enabling traceability and accountability* – traced back to the original designer or manufacturer, which is essential for accountability.
 - *Fostering trust in the market* – enhances trust among stakeholders.

- Properties of hardware watermark

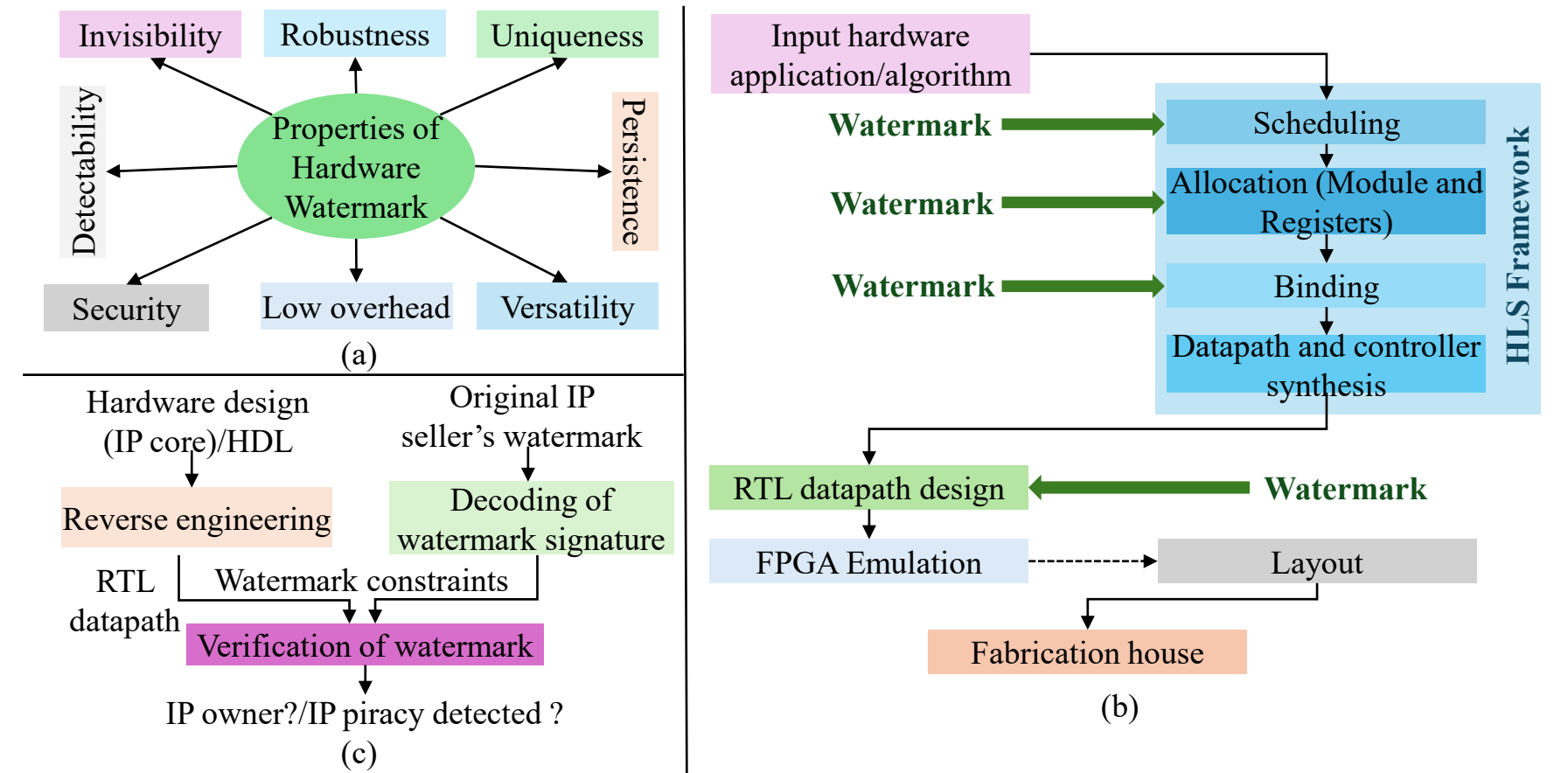


Fig. 1.(a) Different properties of hardware watermark.

(b) Depiction of possible watermark insertion locations during the hardware design process.

(c) Hardware watermark detection process.

- Properties of hardware watermark

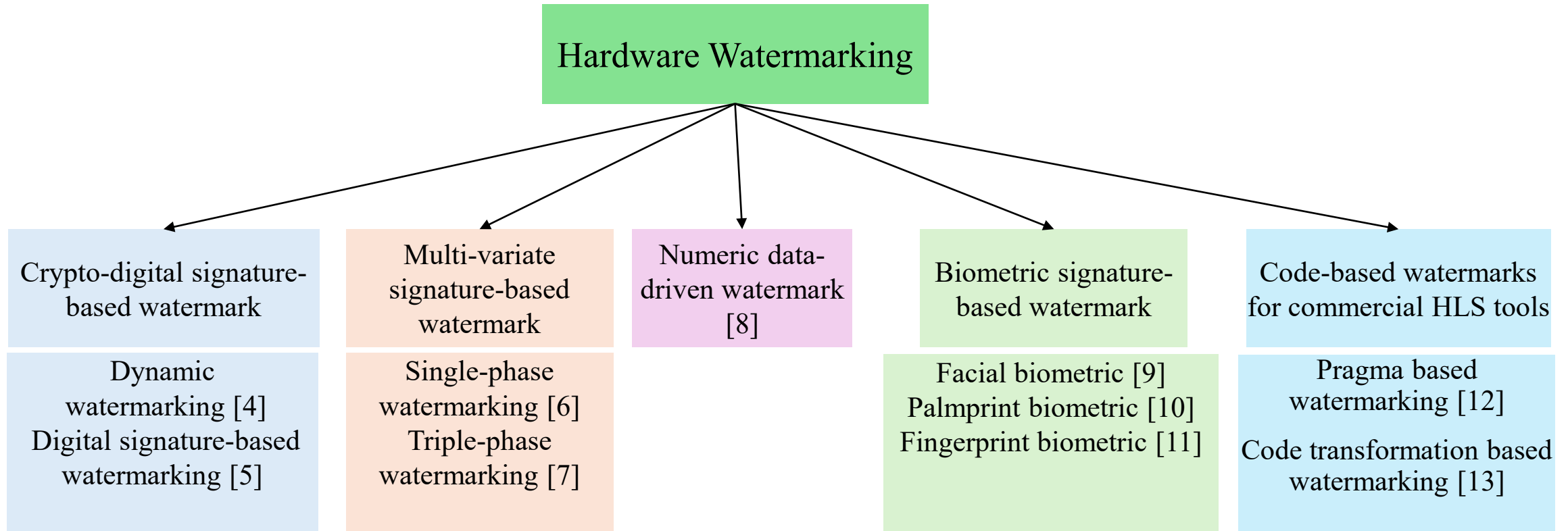


Fig. 2. Taxonomy of HLS-based hardware watermarking approaches.

• Taxonomy of HLS-based hardware watermarking approaches

Table 1
Comparison of HLS-based hardware watermarking approaches based on characteristics/features

Watermarking Approaches	Features/characteristics of watermarking approaches						
	Crypto-logic for signature storage/generation	Multi-variable signature encoding	Usage of CIG for signature embedding	Applicable for commercial HLS tools	Usage of IP vendor/seller biometric	Signature embedding during FU allocation/FU binding/ Scheduling	Tamper tolerance and probability of coincidence analysis
Dynamic watermarking [4], 2005	√	X	√	X	X	X	√
Numeric data-driven watermark [8], 2012	X	X	X	√	X	X	√
Single-phase watermarking [6], 2016	X	√	√	X	X	X	√
Triple-phase watermarking [7], 2017	X	√	√	X	X	√	√
Digital signature-based watermarking [5], 2019	√	X	√	X	X	X	√
Fingerprint biometric based watermarking [11], 2020	√	X	√	X	√	X	√
Pragma based watermarking [12], 2021	X	X	X	√	X	√	X
Code transformation based watermarking [13], 2021	X	X	X	√	X	√	X
Facial biometric based watermarking [9], 2021	√	X	√	X	√	X	√
Palmprint biometric based watermarking [10], 2021	√	X	√	X	√	X	√

• Details of hardware watermarking approaches

- **Key contributions of [4]:** Presents a dynamic hardware watermarking approach using the CIG framework of the HLS process.

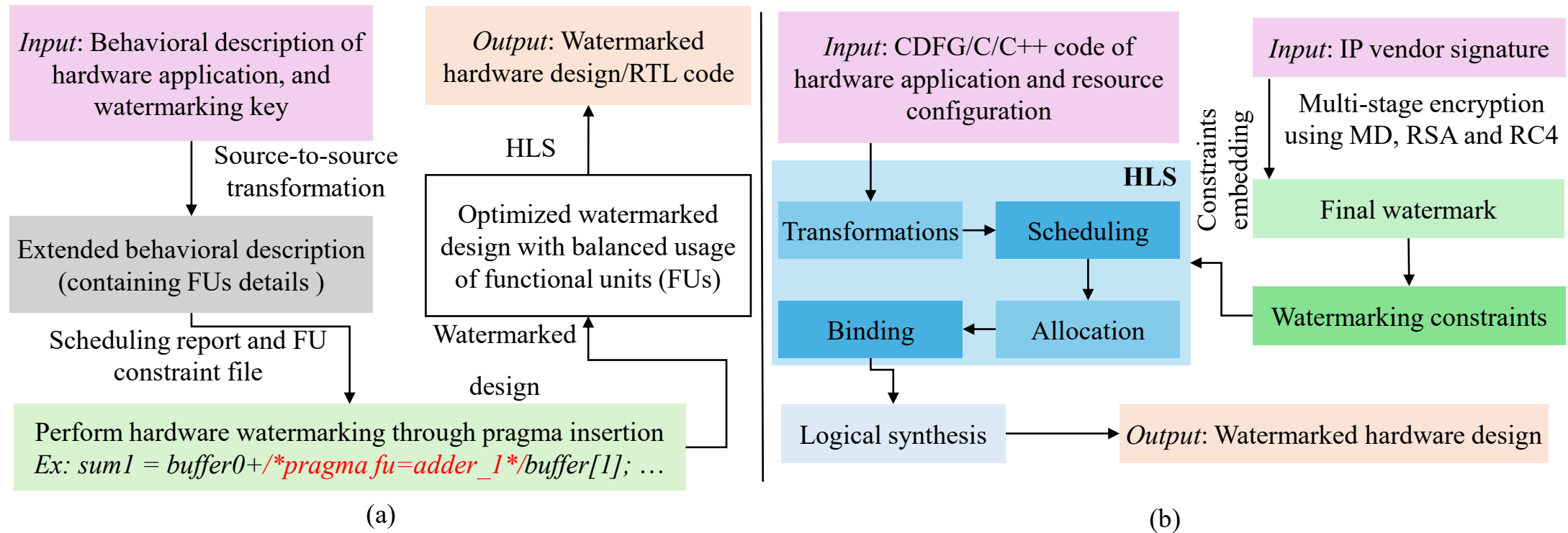


Fig. 3. (a) Details of pragma-based hardware watermarking approach [12], (b). Details of dynamic hardware watermarking approach [4]

• Details of hardware watermarking approaches

- **Key contributions of [5]:** Demonstrates the generation of a robust hardware watermark using SHA-512, RSA cryptosystem, and IP seller-selected encoding rule.

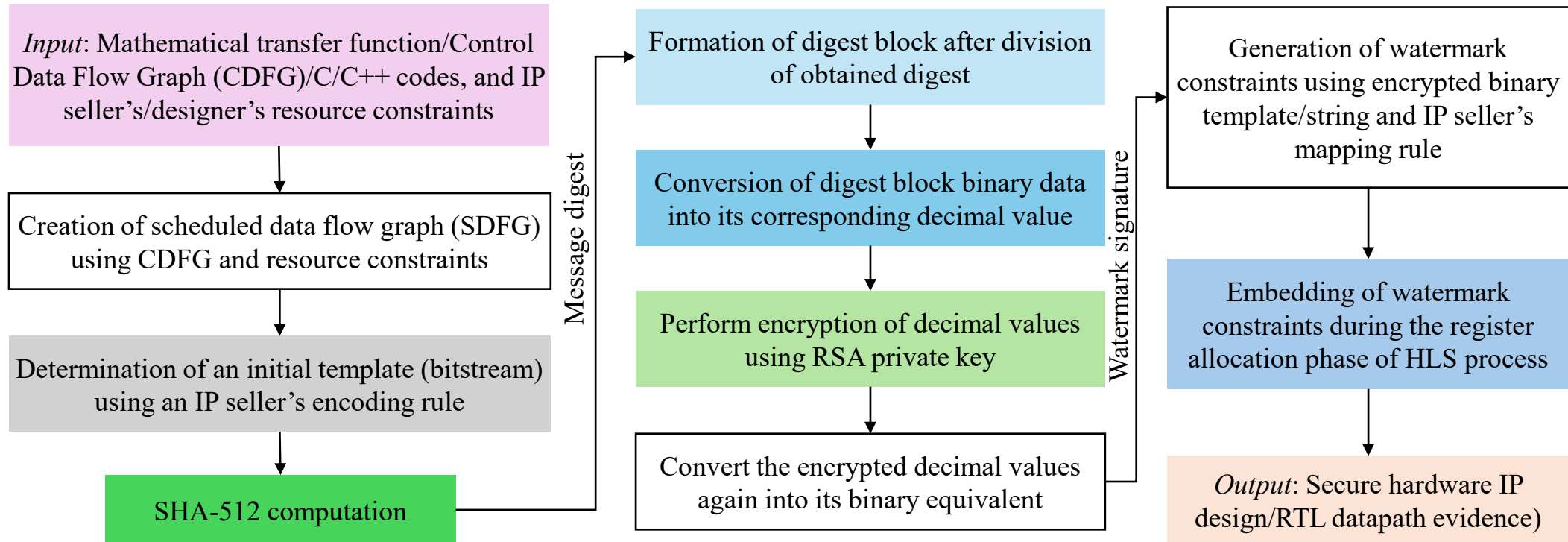


Fig. 4. Details of HLS-digital signature-based watermark generation [5]

• Details of hardware watermarking approaches

- Multivariate signature-based watermark [6], [7]

➤ **Key contributions of [6]:**

- Presents a quadruple variable-based hardware watermarking methodology.
- Exploits the register allocation phase of the HLS

➤ **Key contributions of [7]:**

- Presents a multivariate (using a seven-variable encoding scheme) signature-based hardware watermarking approach.
- Exploits scheduling, hardware allocation, and register allocation phases of the HLS process to embed hardware watermark.

- Details of hardware watermarking approaches

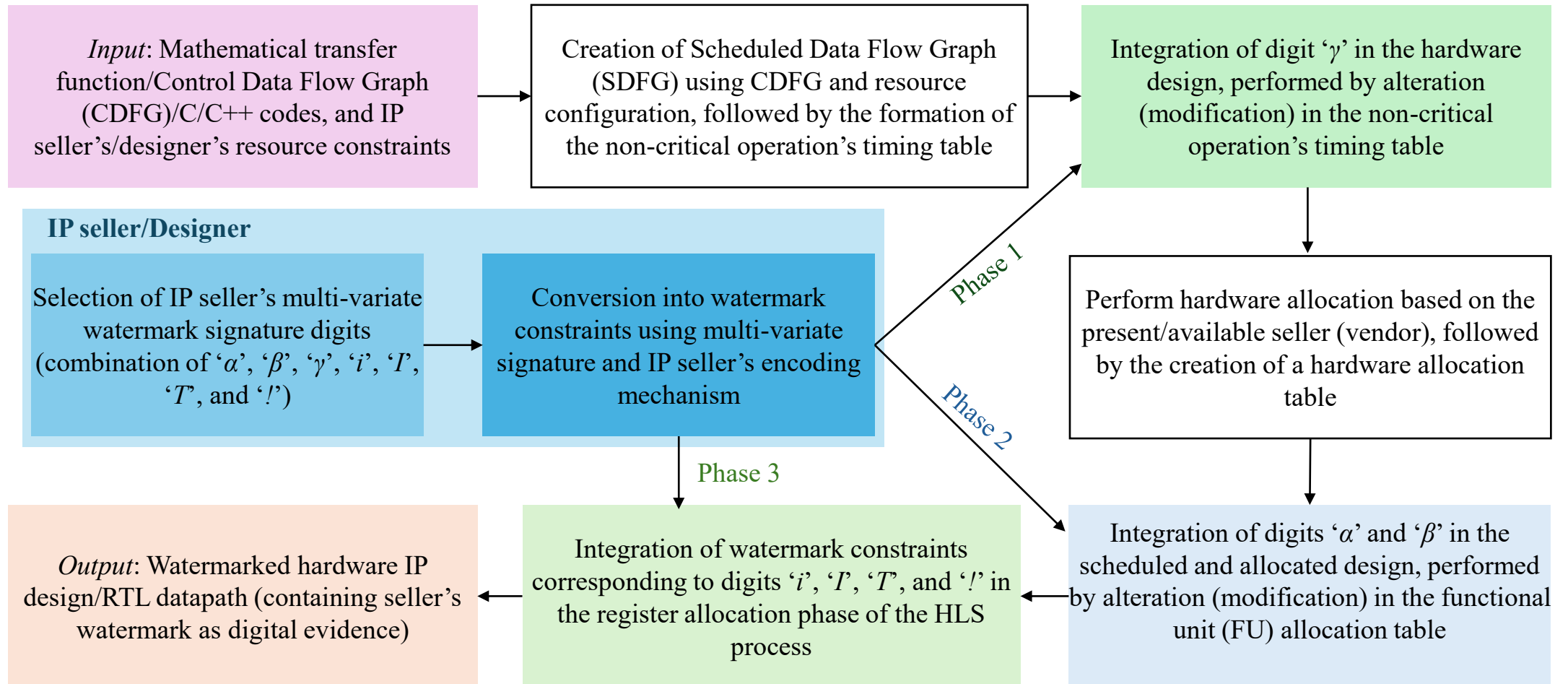


Fig. 4. High level synthesis (HLS) flow depicting the triple-phase hardware watermarking technique [7]

• Details of hardware watermarking approaches

- Numeric data-driven watermark [8]

- **Key contributions of [8]:**

- Presents a numeric data-driven watermarking approach by exploiting mathematical relations between input and output during behavioral synthesis of the HLS process.

- Discusses two different mathematical watermark generation processes:

- 1) a low-cost watermark and
 - 2) a costless watermark.

• Details of hardware watermarking approaches

Code-based watermark for commercial HLS tools [12], [13]

➤ **Key contributions of [12]:**

- Presents a pragma insertion-based hardware watermarking approach using a commercial HLS tool.
- The core watermarking step involves devising a unique FU binding solution, which guarantees that the resulting RTL code is unique.

➤ **Key contributions of [13]:**

- Presents a C-code obfuscation-based watermarking approach to generate a unique hardware birthmark.

- Details of hardware watermarking approaches

Biometric signature-based watermark [9], [10], [11]

- **Key contributions of [9]:**

- Presents a facial biometric-based hardware watermarking methodology using the IP seller's facial biometric characteristics.

- **Key contributions of [10]:**

- Presents a palmprint biometric-based hardware watermarking methodology using the IP seller's palmprint biometric characteristics.

- **Key contributions of [11]:**

- Presents a fingerprint biometric-based hardware watermarking methodology using the IP seller's fingerprint biometric characteristics.

- Details of hardware watermarking approaches

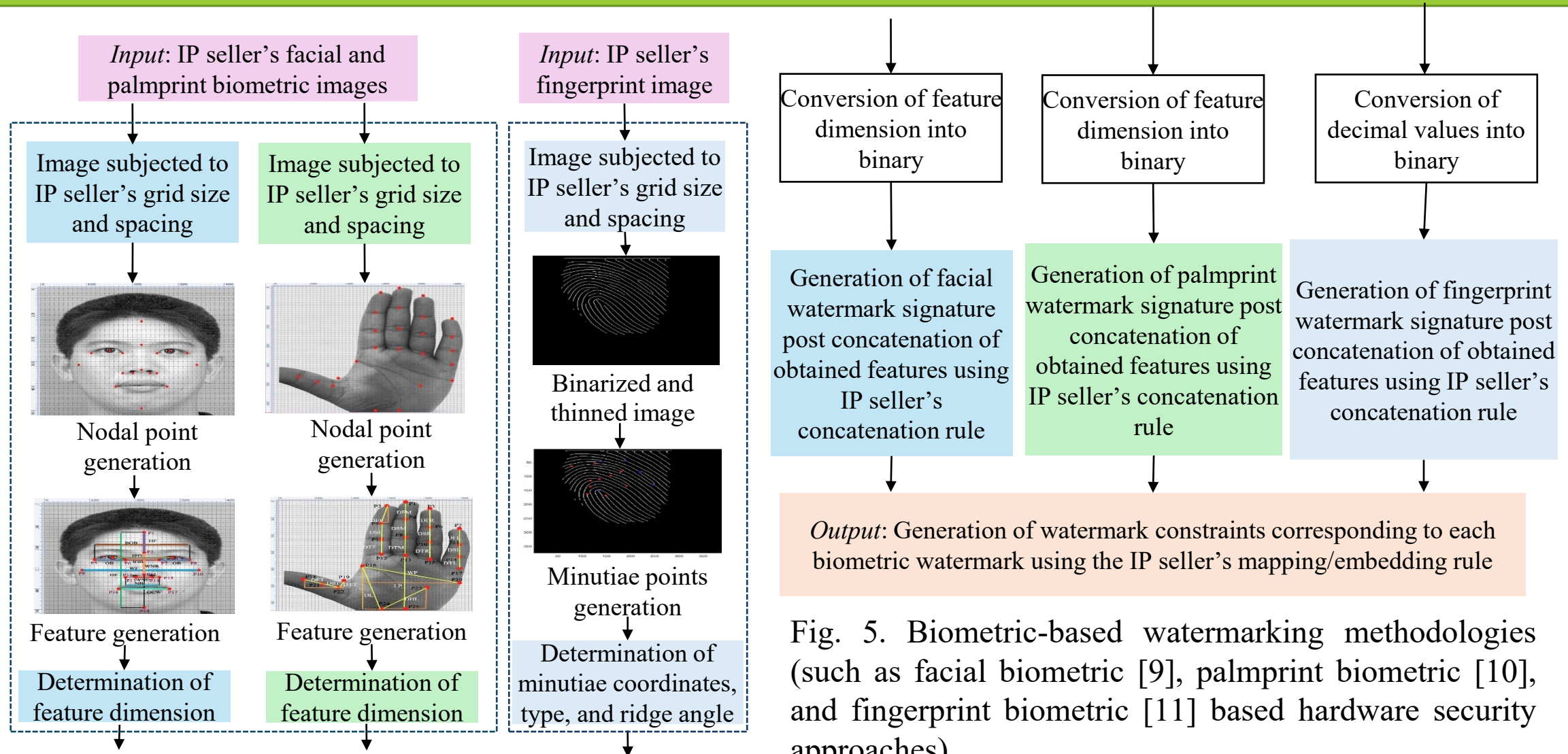


Fig. 5. Biometric-based watermarking methodologies (such as facial biometric [9], palmprint biometric [10], and fingerprint biometric [11] based hardware security approaches)

- Details of hardware watermarking approaches

Table 2. Comparative perspective between facial, palmprint, and fingerprint biometric approach.

S. No.	Parameters	Fingerprint biometric approach [11]	Facial biometric approach [9]	Palmprint biometric approach [10]
1.	Dependence on external factors	Yes; grease and dirt may affect the fingerprint verification process	No	No
2.	Pre-processing	Image enhancement is required for extracting accurate minutiae points	Not required	Not required
3.	Implementation complexity	high	less	Moderate
4.	Security approach	Depends on minutiae points generation on fingerprint image	Depends on nodal points generation on facial image	Depends on nodal points generation on palm image of IP vendor
5.	Probability of coincidence	Lesser Pc value, indicating stronger digital evidence	Pc value higher than fingerprint and palmprint biometric approach	Pc value higher than fingerprint and lower than facial biometric
6.	Tamper tolerance capability	Moderate	Lower than palmprint and fingerprint-based approach	Higher
7.	Digital template regeneration by an adversary	Not possible (as regeneration of digital template depends on minutiae points, feature set, feature order, grid size, etc.)	Not possible (as regeneration of digital template depends on nodal points, feature set, feature order, grid size, etc.)	Not possible (as regeneration of digital template depends on nodal points, feature set, feature order, grid size, etc.)

• Security analysis and discussion

- The probability of coincidence (PC shown in Fig. a) metric evaluates the likelihood of coincidentally identifying the same watermarking constraints in an unsecured design, serving as an indicator of false positives.
- The tamper tolerance (TT value shown in Fig. b) metric assesses the watermarking method's robustness against brute-force and tampering attacks.

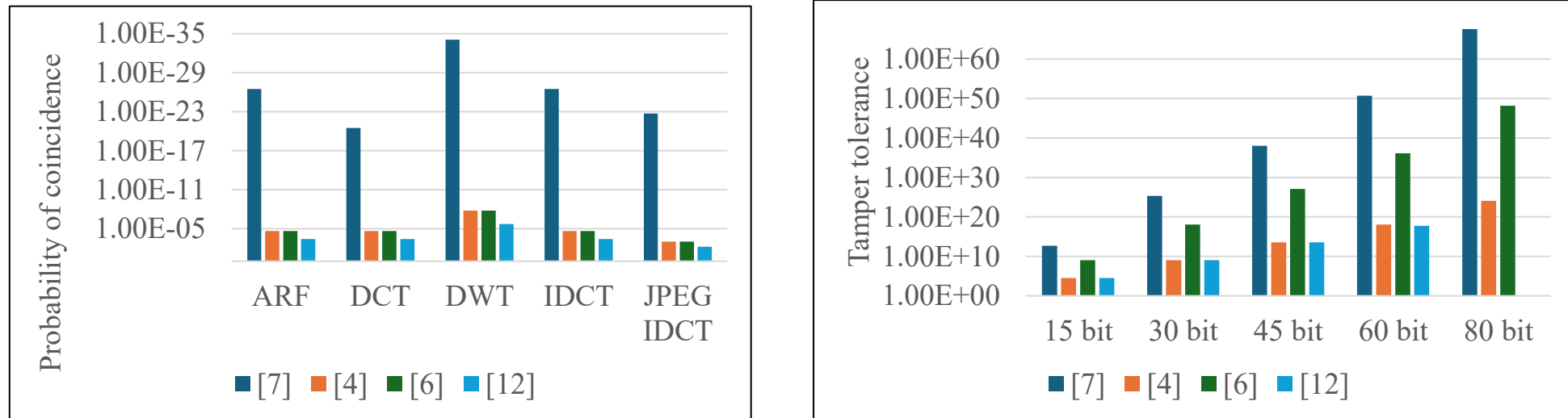
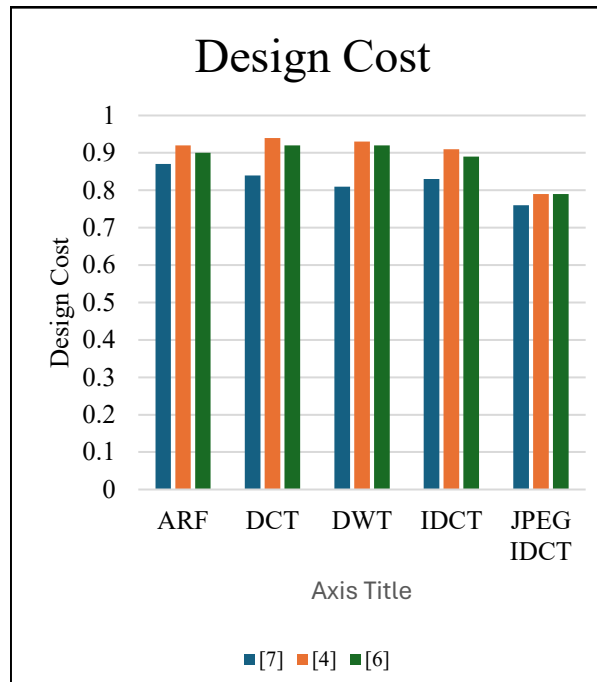
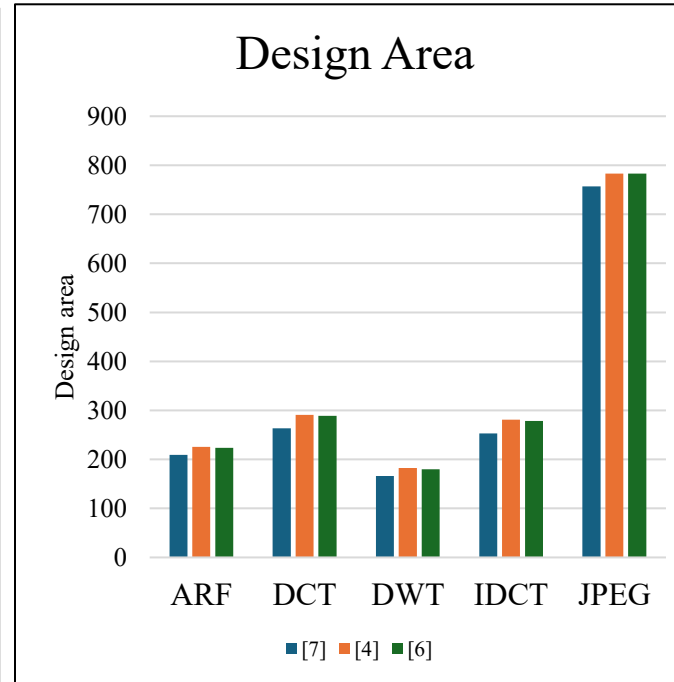


Fig. 6. (a). Comparison of the probability of coincidence among [4], [6], [7], and [12], (b). Comparison of the tamper tolerance among [4], [6], [7], and [12],

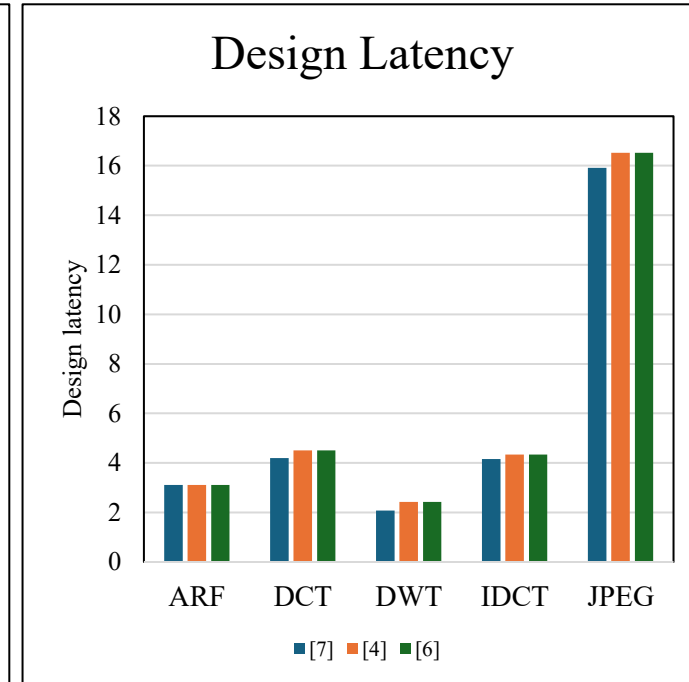
- Design Cost (Area and Latency)



(C)



(d)



(e)

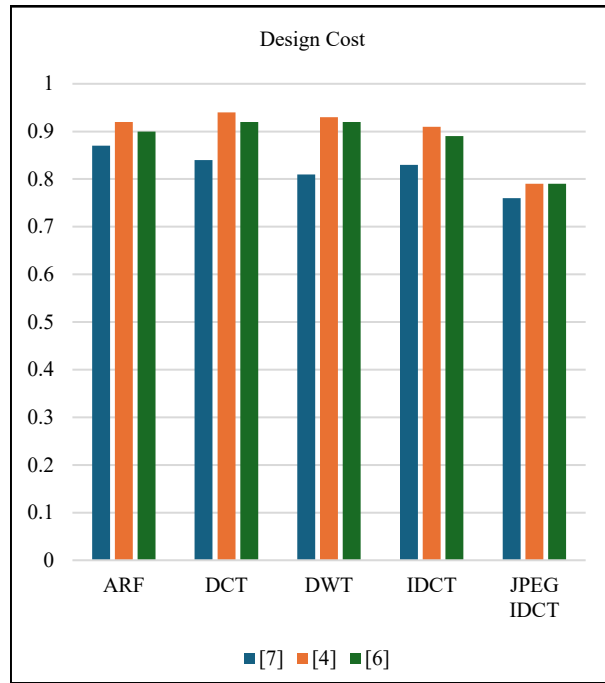
Fig. 6. (c). Comparison of design cost among [4], [6], and [7], (d). Comparison of design area among [4], [6], and [7], and (e). Comparison of design latency among [4], [6], and [7]

[4] F. Koushanfar, I. Hong and M. Potkonjak, "Behavioral synthesis techniques for intellectual property protection," ACM Trans. Des. Autom. Electron. Syst., vol. 10, no. 3, pp. 523-545, 2005.

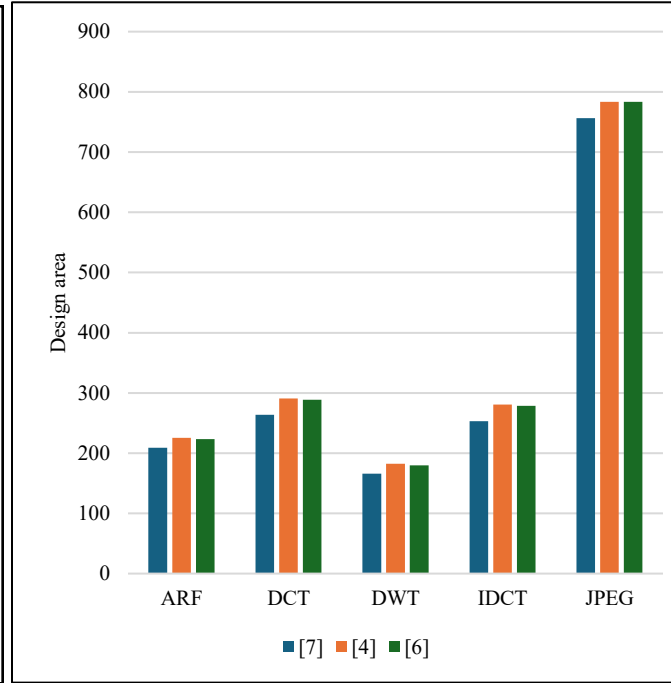
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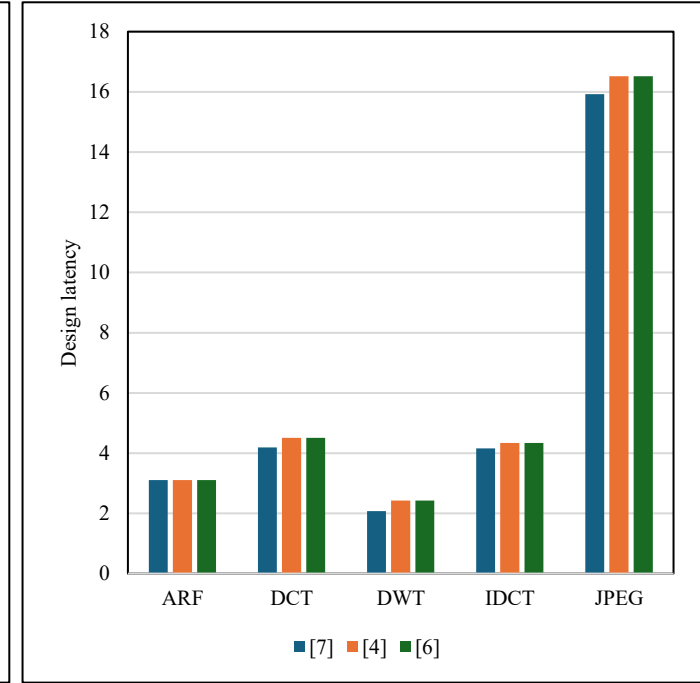
- Design Cost (Area and Latency)



(C)



(d)



(e)

Fig. 6. (c). Comparison of design cost among [4], [6], and [7], (d). Comparison of design area among [4], [6], and [7], and (e). Comparison of design latency among [4], [6], and [7]

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THANK YOU